PAPER_1518 — MAD Disk Efficiency $\eta_{EM} = 1/SO_5^2 = 0.01$ EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket F (AGN / Jet) **Date:** June 18, 2026 **Status:** CLOSED — MAD Poynting-flux electromagnetic efficiency = integer primitive

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Observation

PAPER_817 (GRMHD Binary BH Merger Accretion Modulation) specifies the canonical magnetically-arrested-disk (MAD) Poynting luminosity:

$$L_{Poynt,MAD} \approx \eta_{EM} \cdot \dot{M} \cdot c^2 \qquad \text{with } \eta_{EM} \approx 0.01$$

The factor η_{EM} is widely cited (Tchekhovskoy 2011, Narayan 2003, Yuan & Narayan 2014) as the universal MAD efficiency upper bound.

UQFF Closed Identity

$$\eta_{EM} = 1 / SO_5^2 = 1/100 = 0.01 \quad \text{EXACT}$$

Physical Interpretation

The 1% conversion efficiency of accretion mass-energy to Poynting flux in a MAD-state black hole disk is **not arbitrary** — it equals the inverse-square of the SO_5 integer primitive. The same SO_5 = 10 governing: - $F_{TRZ} = 1/SO_5$ (time-reversal-zone factor) - F_U decay = $1/SO_5^3$ (PAPER_1507) - U_UA coupling = $1/SO_5^4$ (PAPER_1498) - Sun-quiet B field = $1/SO_5^4$ (PAPER_1486) - NS radius = SO_5^4 (PAPER_1513)

…now governs the **electromagnetic efficiency of every magnetically-arrested-disk-state quasar jet** at exactly $1/SO_5^2$.

Predictive Consequence

Any MAD-state quasar jet survey should cluster η_{EM} near 1% with deviations attributable to higher-order F_{TRZ}, β_i corrections. Outliers above 10% suggest non-MAD jet states.

NOT REPLACEMENT

SM/GRMHD treats η_{EM} as an empirical numerical result from simulations. UQFF supplies a structural integer-primitive value matching the same number to two significant figures.

Reference

- Source: PAPER_817 GRMHD Binary BH Merger Accretion Modulation
- Related: PAPER_1009 (3C273), PAPER_1037 (AGN buoyancy), other SO_5-power identities
- Calculator dispatch: `calculate_paradox({"paradox": "mad_efficiency_1_over_so5_2"})`
